Perceived Environmental Uncertainty, Organization Structure and Boundary Spanning Behavior

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ABSTRACT

Boundary spanning activity is conceived of as an intervening variable between organization structure and perceived environmental uncertainty. This is because boundary spanners operate at the skin of the organization and hence their functions are to interpret environmental conditions and relay that information to organization decision makers. Three hypotheses examining relationships between perceived environmental uncertainty, organization structure, and boundary spanning activity were investigated. In a field study comprising 12 work units (182 people) working in a health and welfare organization, positive relationships were found between all three variables. Partial relationships indicated that PEU does not influence the boundary spanning-structure relationship, but structure reduces the PEU-boundary spanning relationship to zero, and boundary spanning reduces the structure-PEU relationship to zero. Thus, not only is the boundary spanning as an intervening variable supported, but results suggest that causality in the structure-PEU relationship may be in the direction of structure to PEU rather than current contingency notions of PEU effecting structure.

INTRODUCTION

This paper reports the results of a field study aimed at delineating the relationships among perceived environmental uncertainty, organization structure, and boundary spanning behavior. Although a number of scholars have addressed the relationship between an organization's environment and its structure (Downey and Slocum (8), Darran, Miles, and Snow (6), Thompson (25)), there have apparently been few studies of the possibility that these variables or this relationship might be related to the nature of the boundary spanning behavior of the organization's members.

This void in the empirical literature is unfortunate since, although the circumscribing or limiting nature of an organization's boundary is important (cf. Child (5), Evan (11), Levine and White(17), and Utterback (25)), so also is the role of boundaries as regulators of the flow of information between the organization and its environment (cf. Miller (21), Miller and Rice (20), Organ (22), and Thompson (25)). In view of the potential importance of this boundary-crossing information flow to a more complete theory of organizations and their environment, this study was designed to determine whether the frequency of boundary spanning activity was related to either perceived environmental uncertainty or organization structure.

The feature of the environment dealt with in the study was its uncertainty. In particular, the study was concerned with perceived environmental uncertainty, the uncertainty in the environment that is perceived by the organizational member. The conclusion of Miles, Snow, and Pfeffer (19) in particular suggests the appropriateness of using this variable in empirical studies of open systems or contingency theory. These researchers found that, within the same objective environment, there were organizations (college textbook publishers) whose top managers perceived little or no change and uncertainty in the environment as well as organizations whose top managers perceived continuous change and uncertainty in the environment. They then noted (p. 262) that these two types of perceptions, and hence the organizational structures that were associated with them, can survive and even flourish, at least in the short run. This reasoning, the review and conceptual work of Downey and Slocum (8), and Weick's (27) notion of an "enacted environment" all suggest that the organizational design actions an organization takes in responding to its environment may well be more consistent with perceptions of the environment than with more objective indicators of environmental conditions. In this study, the concept of organizational environment was operationalized as perceived environment uncertainty (PEU).

In particular, the study was directed at two related issues. The first of these concerns the possibility that since contingency theory as the term is currently used in discussions of the organization-environment interface (cf. 26 and 13) includes the concept of information passing from the environment to organizational designers, is it not reasonable to include in the theory a measure of the intensity of information gathering activity at the boundary of the organization? The second issue concerns the nature of whatever linkages might exist between organizational structure and PEU. Is it possible, for example, that reported correlations between structure and PEU might be largely a consequence of (1) a correlation between internal structure and boundary spanning activity and (2) a correlation between boundary spanning activity and PEU? These two issues were addressed in this study.

HYPOTHESES

Three hypotheses were tested in this study, each one describing the relationship between two of the three variables studied. As will be seen, each hypothesis was tested using both first and second order analyses.

The first hypothesis concerns the relationship between perceived environmental uncertainty and the nature of the organizational structure. Contingency theory suggests that PEU affects organizational structure, that administrators design organizational structures such that the organization will be able to more effectively respond to environmental demands and that they do this in accord with their perceptions of the environment. In general, it is argued that the more uncertainty that is perceived, the more will be the looseness of flexibility or "organicness"(Burns and Stalker, 4) of the organizational structure, i.e., that a positive relationship exists between these variables. For example, on the basis of a number of field studies Osborn and Hunt (22) concluded that either perceived or objective environmental uncertainty is a determinant of organizational structure. In particular they stated that "The work of Burns and Stalker (1961), Chandler (1962), Emery and Trist (1965), Lawrence and Lorsch (1967), and Neghandi and Reiman (1973), among others, has indicated that as [the task environment] becomes more dynamic, the organization must become not only more receptive to change, but alter its internal structure and operations to maintain and/or establish a high survival potential" (p. 232). The field studies of Duncan (9) and Khandwalla (15) seem to support this PEU-influences-structure argument, and McWhinney (18) used it in proposing various decision making modalities appropriate to the types of environments described by Emery and Trist (10). On the other hand, if organizational structures have as a function the processing of information for reducing uncertainty, then structure would certainly affect PEU. A recent experimental study by Huber, O'Connell and Cummings (13) supports this structure-influences-PEU argument. On the basis of either or both of these arguments, we hypothesized the following. HYPOTHESIS 1: ORGANICNESS OF STRUCTURE WILL BE POSITIVELY ASSOCIATED WITH PERCEIVED ENVIRONMENTAL UNCERTAINTY.

The second hypothesis concerns the relationship between the nature of the boundary spanning activity (as defined below) and the nature of the organizational structure. This relationship is relatively unstudied, except for the work of Aiken and Hage (2) and Brown (3). The former hypothesized and found that "... the greater the degree of organizational permeability, i.e., the greater the extensity of boundary spanning roles and the greater the number of interorganizational linkages, the greater the internal structure of the focal organization will tend toward the organic (Burn and Stalker, (4)) or dynamic (Aiken and Hage, (1)) organizational style" (p. 5). Brown's reasoning was of a similar nature, and had to do with the function of boundaries and organizational level. Our own reasoning extends these arguments. We feel that in uncertain environments, organizations will have a high need to: (1) obtain information, e.g. to have a high frequency of boundary spanning activity; and (2) process it in a way that will maximize the number of points of view to which it is exposed, e.g. to create an organic structure, and that in more certain environments, organizations will have a lower need to do either. If these arguments hold, then it is clear that boundary spanning and organicness would be associated with one another. As a consequence, then, of this reasoning and the admittedly sparse literature on the subject, we hypothesized the following. HYPOTHESIS 2: ORGANICNESS WILL BE POSITIVELY ASSOCIATED WITH FREQUENCY OF BOUNDARY SPANNING ACTIVITY.

The third hypothesis concerns the relationship between perceived environmental uncertainty and boundary spanning activity. Because the relationships of the first two hypotheses were predicted to be positive, it follows that we must hypothesize a positive relationship between these two variables. It is important to note, however, that a positive relationship may be reasonable to posit in any case. That is, if organizational members attempt to reduce perceived environmental uncertainty by obtaining more information, it follows that those perceiving a high degree of environmental uncertainty would engage in more boundary spanning activity in order to bring the uncertainty down to some manageable level. Similarly, in environments that are not perceived as uncertain, where the perceived uncertainty is already at some quite manageable level, we would expect to find low perceived uncertainty and also less boundary spanning activity. The paucity of empirical work with which to bolster these arguments was, of course, a motivating force to conduct the study reported here. Combining a reliance on these arguments with the logical necessity of expecting a positive association between two variables that are each assumed to be positively associated with a third, we hypothesized the following. HYPOTHESIS 3: FREQUENCY OF BOUNDARY SPANNING ACTIVITY WILL BE POSITIVELY ASSOCIATED WITH PERCEIVED ENVIRONMENTAL UNCERTAINTY.

METHODOLOGY

Research Setting and Data Collection Procedures

This cross sectional field study utilized a health and welfare organization (H&W), concerned with family problems, adoption, social work, etc., in a midwestern state government. The subjects were 220 professional and clerical employees in twelve work groups. All members of the sampled work groups were tested. Each questionnaire was coded to provide anonymity of respondents. A member of H&W assigned a number to each name. Completed questionnaires were given directly to the research team. Thus the agency did not see individual results; the researcher did not see the names. The questionnaire was administered to groups of no more than 15. The organization assigned personnel to testing sessions and kept track of absences. Follow-up testing of non-respondents consisted of giving each non-respondent a questionnaire to be returned by mail. One hundred and eighty-two usable questionnaires were obtained for a response rate of 83%.

These personnel work in the lowest two echelons of a tall hierarchy which means these results may generalize only to lowest levels of organizations. However, as Brown (3) suggests, the technical level (22), involving the acquisition and utilization of technical knowledge, involves important boundary spanning functions. This operational information is primarily obtained for use with organizational control functions (3, p. 325) and as such has an impact on decision making activities of those in higher echelons, as well as technical level decision making.

Measures of the Variables

Three composite variables were used in the field study: organicness, frequency of boundary spanning activity, and perceived environmental uncertainty.

Six measures were included in the organicness variable. Five of these, (1) participation in work decisions, (2) formalization (reverse scored), (3) hierarchy of authority (reverse scored), (4) impersonality (reverse scored) and (5) division of labor measures were taken from Duncan (8). The sixth measure was a questionnaire item directed at ascertaining the extent of the subject's participation in strategic decisions. A factor analysis indicated that each of the six variables loaded on the first factor with a loading of from .54 to .86 which accounted for twice the variance of the second factor which was a participation--authority vs. formalization--division of labor factor. Mean scores for each unit were calculated by finding the mean of the individual responses on each measure and then these six unweighted scores were then added together and their mean used as the organicness score for the respective unit. Leifer (16) presents a rationale for labeling this index "organicness."

Seven questionnaire items were used in a composite measure of perceived environmental uncertainty. An example item is "How often do you believe that the information you have about factors outside the Division of Family Services is sufficient for decision making?" The responses were collected on a 0-100 scale anchored at five points, never, seldom, sometimes, often, always. These seven items are discussed in detail in Leifer (16). A factor analysis indicated that each question loaded on the first factor with a loading of from .37 to .74. A second factor distinguished overall perceptions of the environment from informational uncertainty with respect to decision making. The seven unweighted scores were added together and their mean used as the perceived environmental uncertainty score for the respective individual.

Eight questions concerning the subject's frequency of boundary spanning activity were asked (see Appendix)--one question applicable to each of the eight cells shown in Table 1. A factor analysis indicated that each of the eight questions loaded on the first factor (which accounted for twice the variance of the second) with a loading of from .44 to .83. The second factor distinguished verbal and written activities. Test-retest reliability was .76. These eight unweighted scores were added together and their mean used as the boundary spanning activity score for the respective individual.

It should be noted that this composite measure of boundary spanning activity, based as it is on behavior, is quite far removed from the effects of organizational structure as operationalized in this study. Inspection of the eight activity measures and the six organicness measures does not suggest that the frequency of the eight boundary spanning behaviors would be determined by the overall structural attributes of the organization of which the individual was a member. In addition, whatever linkage might exist between individual perceptions of organicness and individual boundary spanning behaviors would have had little impact in this study, since the organicness score used was the mean of the scores obtained from all members of the individual's work unit while the boundary spanning score was strictly a function of the individual's personal activity level.

TABLE 1

Measures of Boundary Spanning Activity

	Between Organizations But Within the Larger Institution		Outside the Larger Institution Entirely	
	Formal	Informal	Formal	Informal
Written	Q5	Q7	Q7	Q8
Verbal	Q1	Q3	Q2	Q4

Data Analysis

The unit of analysis was the individual subject. The 182 scores were organized in three contingency tables with approximately equal numbers of subjects in the marginals, as shown in Tables 2, 3, and 4. The data were first analyzed using the Goodman-Kruskal Gamma (γ), (10) a nonparametric measure of association yielding values of $-1 \leq \gamma \leq +1$. This measure has an interpretation similar to that of a correlation coefficient, but is applicable to ordinal scale and contingency table analysis. We felt that interval scale interpretation of these perceptual measures was not appropriate and hence the nonparametric statistic. The data were then analyzed using a partial gamma as in Davis (5).

Table 2*

Relationship Between Organicness and Perceived Environmental Uncertainty

	Perceived Environmental Uncertainty				
	34 and less	34-41	41-50	50-100	
Organicness					
Low	29	27	24	24	104
High	14	17	23	24	78
Total	43	44	47	48	182

$\gamma = .21$ ($p < .05$)

TABLE 3*

Relationship Between Boundary Spanning Activity (Permeability) and Organicness

	Boundary Spanning Activity				
	10 and less	10-28	28-62	62-404	
Organicness					
Low	39	29	21	15	104
High	5	15	25	33	78
Total	44	44	46	48	

$\gamma = .63$ ($p < .01$)

*These data reflect approximately equal numbers of work groups in each category, but since there are unequal numbers of people in each work group, there are unequal N.

TABLE 4

Relationship Between Boundary Spanning Activity (Permeability) and Perceived Environmental Uncertainty

	Perceived Environmental Uncertainty				
	34 and less	34-44	41-50	50-100	
Boundary Spanning Activity					
10 and less	19	9	12	4	44
10-28	11	11	10	12	44
28-62	6	12	11	17	46
62-404	7	12	14	15	48
Total	43	44	47	48	

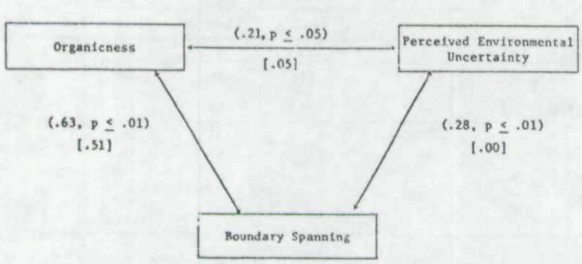
$\gamma = .28$ ($p < .01$)

RESULTS

The data for testing the three hypotheses are shown in Tables 2,3, and 4. The three first-order γ values associated with the three pairings of variables are shown in parentheses in Figure 1. Since all three values indicate positive associations between the respective variables (after adjusting for the reverse scoring on two of the variables), it is possible that the sign and magnitude of any one value could be simply the consequence of the sign and magnitude of the other two. In order to determine whether this were true, or if there really were underlying associations between the variables of each pairing, the second order were calculated and are shown within the square brackets of Figure 1.

FIGURE 1

Relationships Between Boundary Spanning, Perceived Environmental Uncertainty, and Organicness



Hypothesis 1 states that there will be a positive relationship between organicness and PEU. From Table 2 and Figure 1 we see that the observed relationship [γ=.21] was positive and marginally significant ($p<.05$), but did not exist [γ=.05] after the variance due to boundary spanning activity was removed. When only those individuals scoring in the upper quartile on the boundary spanning measures were included, the PEU-organicness relationship was γ=.36 ($p<.01$), whereas when only those scoring in the lower three quartiles were included, γ was nonsignificant. These facts suggest that, at least at the boundary of the organization, the strong association between organic structure and high PEU occurs primarily if not solely under the condition of high information processing activity.

Thus the hypothesis was supported subject to the strong qualification that the positive relationship appears to be a consequence of the relationships that exist between the two variables of the hypothesis and the level of boundary spanning behavior. This finding lends support to the notion that boundary spanning may impact the relationship between PEU and structure and should be made a more explicit and studied component of contingency theory.

Hypothesis 2 states that there will be a positive relationship between organicness and boundary spanning activity. From Table 3 and Figure 1 we see that the hypothesis was supported. The observed relationship (γ=.63) was positive and highly significant ($p<.01$) and existed (γ=.51) even after the variance due to perceived environmental uncertainty was removed. Thus the hypothesis was supported without qualification.

Hypothesis 3 states that there will be a positive relationship between boundary spanning activity and perceived environmental uncertainty. From Table 4 and Figure 1 we see that the observed relationship (γ=.28) was positive and highly significant ($p<.01$), but did not exist [γ=.00] after the variance due to organicness was removed. Thus this hypothesis was supported subject to the strong qualification that the positive relationship appears to be a consequence of the relationships that exist between the two variables of the hypothesis and the organicness of the organization of which the individual is a member.

SUMMARY AND DISCUSSION

One of the issues of interest in this study was the possibility that boundary spanning activity might be usefully included as a component variable in an expanded form of contingency theory. The results reported here suggest that this is, in fact, the case. In particular, the test of Hypothesis 1 suggests that the frequency of boundary spanning activity may mediate the association between PEU and structure.

The second issue addressed was the nature of the linkages among PEU, organicness, and frequency of boundary spanning activity. Of considerable interest is the finding that organicness and the frequency of boundary spanning activity were quite strongly associated. Recall that this is not an association between specific role definitions (a measure of structure) and adherence to them (a measure of behavior), an association that we would expect, in general, to be quite strong. Rather we are seeing the relationship between an organizational attribute, organicness, and an individual behavior, communication frequency with external agents, a behavior that might be influenced by many factors, including PEU. Our reasoning, in fact, was that the association between these two variables was a consequence of PEU, that both would increase as PEU increased. The research results do not support this reasoning, as partialling out the effect of PEU reduced the association very little.

The analysis performed here suggests that, even though PEU does not determine the association between organicness and boundary spanning activity and even though the measures used to operationalize these variables do not necessarily imply an association, a strong relationship exists. Certainly more research on this matter is in order.

Pursuing further the issue of the nature and direction of the linkages among these three variables, we note that the findings reported here suggest that boundary spanning activity intervenes between and mediates the relationship between perceived environmental uncertainty and organization structure. Inasmuch as most studies in the literature relating structure to perceived environmental uncertainty utilize zero order correlations, it has been logical for researchers to posit the PEU-influences-structure (or outside in) reasoning that drives from an open systems theory of organizations.

Findings from this study suggest an alternative to this position. The relationship between PEU and organicness disappeared when the frequency of boundary spanning was partialled out (hypothesis 1). From a test of Hypothesis 3, we saw that when organicness was partialled out of the relationship between PEU and boundary spanning activity, the relationship again disappeared. When PEU was partialled out of the structure-boundary spanning activity relationship, the relationship was not appreciably reduced. It appears that, on the basis of these results, structure influences boundary spanning activity to a greater degree than PEU.. In addition, it appears that boundary spanning mediates the relationship between structure and PEU. That is, boundary spanning activity may determine the amount of perceived environmental uncertainty (see results of hypothesis 2) as a function of and not a cause of the information processing capabilities of structure. This implies that inasmuch as structure is an administratively controlled variable, the amount of PEU admitted at the boundary of the organization is also administratively controllable. These results are still complementary to previously found correlational relationships between structure and PEU only an inside out or structure-influences-PEU reasoning for the relationship is suggested.

This reasoning suggest that an interaction between internal organization factors and external environmental factors produces both PEU and organization structure. Since the variables are both perception, it is still not clear at what point perceptions are tracking actual external events or internally held individual biases.

It should be noted that the results of this study pertain only to the lowest levels of the fairly hierarchical and large institution studied. Certainly further research on the interrelationships among these variables for middle and executive officers or organizations is needed. Nevertheless, the research results reported here suggests that a theory purporting to explain the relationship between organizational structure and environment will be more successful in achieving its purpose if it includes as a component variable the frequency of boundary spanning activity.

APPENDIX

MEASURES OF PERCEIVED ENVIRONMENTAL UNCERTAINTY

never seldom sometimes often always

2. How often are there changes in the social, economic, or political conditions outside your work group which directly affect your work?
0 20 40 60 80 100

3. People can often point to prevailing ideas in their profession about the best methods or techniques to be used in their jobs. How often are there changes in such ideas regarding your work?
0 20 40 60 80 100

8. How often do you believe that the information you have about factors outside the Division of Family Services is sufficient for decision making?
0 20 40 60 80 100

9. How often do you need to gather information from outside the Division of Family Services to solve a problem?
0 20 40 60 80 100

10. How often do you feel you are unable to predict the outcomes of your decisions?
0 20 40 60 80 100

47. ...do you know what to expect in your dealings with other people or organizations?
0 20 40 60 80 100

48. ...must you decide anew which person in another organization it's best to talk to?
0 20 40 ·) 80 100

MEASURES OF BOUNDARY SPANNING ACTIVITIES

I. First, we would like to ask you about the contacts you have with members of other organizations, work groups, associations, clients, etc., whether directly in person, or indirectly by telephone or in writing. If you never contact persons outside the Div. of Family Services or your work unit please check the category "not applicable" for the following questions.

1. Excluding professional or occupational associations, in an average week in the performance of your job,

1a. how many hours do you attend formal meetings, (committees, planning, etc.) (If not applicable, check here_____).

a. outside your work group but within Family Services(includes regional ofcs
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 ____How many more

b. outside of Family Services altogether (includes clients)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 ____How many more

1b. how many hours do you confer about your work in informal face-to-face conversation or telephone conversation with people of other organizations. This refers to informal and verbal contacts only. (If not applicable, check here____).

a. outside your work group but within Family Services(includes regional ofcs
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 ____How many more

b. outside of Family Services altogether (includes clients)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 ____ How many more

1c. how many times do you send or receive formal, written, but work related communications in the form of reports or data from people in other organizations. Please include monthly or similarly scheduled reports in this response. (If not applicable, check here ____).

a. outside your work group but within Family Services(includes regional ofcs
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 ____ How many more

b. outside of Family Services altogether (includes clients)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 ____ How many more

1d. how many times do you send or receive informal, written communications in the form of reports, memoes or the like from people in other organizations in regard to your work. (If not applicable, check here _____).

a. outside your work group but within Family Services(includes regional ofcs
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 ____ How many more

b. outside of Family Services altogether (includes clients)
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 ____ How many more

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